Fluid Mechanics

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Q1. A liquid drop of diameter 2 mm breaks into 512 identical droplets. The change in surface energy is $\alpha \times 10^{-6}$ J. The value of α is _____. (Take surface tension of liquid = 0.08 N/m)

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(1) 10
(2) 7
(3) 8
(4) 11

✓ Answer: (2) 7

Solution:

Volume conservation: $R^3 = 512r^3$, so $R = 8r$.
With $R = 1$ mm $= 10^{-3}$ m, $r = 1.25\times10^{-4}$ m.
$\Delta E = 4\pi T(512r^2 - R^2) = 4\pi(0.08)(8.0\times10^{-6} - 1.0\times10^{-6})$
$= 4\pi(0.08)(7.0\times10^{-6}) \approx 7.04\times10^{-6}$ J. So $\alpha \approx 7$.

Q2. A tub is filled with water and a wooden cube 10 cm × 10 cm × 10 cm is placed in the water, floating with part of it submerged. When a metal coin is placed on the wooden cube, the submerged part increases by 3.87 cm. The mass of the metal coin is _____ gram. (Take water density as 1 g/cm³ and density of wood as 0.4 g/cm³)

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✓ Answer: 387

Solution:

Cross-sectional area of cube = $10\times10 = 100$ cm².
Additional submerged volume = $100\times3.87 = 387$ cm³.
This additional buoyant force exactly supports the coin's weight:
mass of coin = (additional volume) × (water density) = $387\times1 = 387$ g.

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in